

LAB ASSIGNMENT – 14.2

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BATCH – 29

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS

Generation

Lab Objectives

- Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.
- Apply responsive, accessible, and interactive design principles.

Week 7 -

Tuesday

- Create practical UI components for real-world web applications.
- Use AI to optimize layout, UX, and performance.

Task Description 1 – (Landing Page Design)

- Create a simple landing page for a product or service. Include a header, hero section, and footer.

Requirements:

- Use HTML and CSS to structure the page.
- Use AI to suggest layout alignment and typography.
- Ensure the page is responsive across devices.

Task Description 2 – (Theme Toggle Feature)

- Provide a button to switch between light and dark themes..

Requirements:

- Use JavaScript to toggle CSS classes.
- Use AI assistance to manage color contrast and accessibility.
- Ensure theme preference applies instantly on click.

Task Description 3 – (User Feedback Form)

- Create a feedback form with fields: Username, Rating, Comments.

Requirements:

- Use JavaScript to validate required fields.
- Display success or error messages dynamically.
- Use AI to improve form usability and layout.

Task Description - 4 (Image Gallery with JavaScript)

- Task: Display a small image gallery on a web page.

Requirements:

- Use JavaScript to dynamically load images.
- Add next/previous navigation buttons.
- Use AI to optimize layout responsiveness.

Task Description - 5: (Notification Banner)

- Display a notification banner at the top of the page.

Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use AI to ensure smooth transitions and clean UI behavior.

PROMPT: Generate a modern responsive web application UI using HTML, CSS, and JavaScript with a navigation bar, hero section, feedback form, image gallery, and notification banner. Include dark mode toggle, smooth UI interactions, and clean card-based layout for better user experience.

COMBINED CODE:

```
from flask import Flask, render_template_string
```

```
app = Flask(__name__)
```

```
html = """
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Premium UI App</title>
```

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

<style>

/* RESET */

```
{  
  margin:0;  
  padding:0;  
  box-sizing:border-box;  
  font-family: 'Poppins', sans-serif;  
}
```

/* BODY */

```
body{  
  background:#f9fafb;  
  color:#111;  
  transition:0.4s;  
}
```

/* DARK MODE */

```
.dark{  
  background:#0f172a;  
  color:white;  
}
```

/* NAVBAR */

```
nav{  
  display:flex;  
  justify-content:space-between;  
  padding:20px 40px;  
  background:white;  
  box-shadow:0 2px 10px rgba(0,0,0,0.05);  
}
```

```
}
```

```
.dark nav{  
  background:#1e293b;  
}
```

```
/* BUTTON */
```

```
.btn{  
  padding:8px 15px;  
  border:none;  
  border-radius:20px;  
  background:#6366f1;  
  color:white;  
  cursor:pointer;  
}
```

```
/* HERO */
```

```
.hero{  
  text-align:center;  
  padding:80px 20px;  
}
```

```
.hero h1{  
  font-size:42px;  
}
```

```
.hero p{  
  color:gray;  
  margin-top:10px;  
}
```

```
/* CARD */

.container{
  display:flex;
  flex-wrap:wrap;
  justify-content:center;
}

.card{
  background:white;
  padding:20px;
  margin:20px;
  border-radius:15px;
  width:300px;
  box-shadow:0 5px 20px rgba(0,0,0,0.1);
  transition:0.3s;
}

.dark .card{
  background:#1e293b;
}

.card:hover{
  transform:translateY(-5px);
}

/* INPUT */

input, textarea{
  width:100%;
  padding:10px;
  margin:10px 0;
  border-radius:8px;
```

```
    border:1px solid #ddd;
}
```

```
/* GALLERY */
```

```
.gallery img{
    width:100%;
    border-radius:10px;
}
```

```
/* BANNER */
```

```
#banner{
    background:#22c55e;
    color:white;
    text-align:center;
    padding:10px;
}
```

```
/* FOOTER */
```

```
footer{
    text-align:center;
    padding:20px;
    color:gray;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- Banner -->
```

```
<div id="banner">
```

🎉 Limited Time Offer!

<button onclick="closeBanner()">X</button>

</div>

<!-- Navbar -->

<nav>

<h2>PremiumApp</h2>

<button class="btn" onclick="toggleTheme()">Toggle</button>

</nav>

<!-- Hero -->

<div class="hero">

<h1>Build Smart Web Apps</h1>

<p>Clean UI + Powerful Features</p>

</div>

<!-- Cards -->

<div class="container">

<!-- Form -->

<div class="card">

<h3>Feedback</h3>

<form onsubmit="return validateForm()">

<input type="text" id="name" placeholder="Username">

<input type="number" id="rating" placeholder="Rating">

<textarea id="comments" placeholder="Comments"></textarea>

<button class="btn">Submit</button>

<p id="msg"></p>

</form>

</div>

```
<!-- Gallery -->
<div class="card gallery">
  <h3>Gallery</h3>
  <img id="img">
  <br><br>
  <button class="btn" onclick="prev()">Prev</button>
  <button class="btn" onclick="next()">Next</button>
</div>
```

```
</div>
```

```
<!-- Footer -->
<footer>
  © 2026 Premium UI | All rights reserved
</footer>
```

```
<script>
```

```
/* THEME */
```

```
function toggleTheme(){
  document.body.classList.toggle("dark");
}
```

```
/* FORM */
```

```
function validateForm(){
  let name=document.getElementById("name").value;
  let rating=document.getElementById("rating").value;

  if(name=="" || rating==""){
    document.getElementById("msg").innerText="Fill all fields!";
    return false;
  }
}
```



```
}

document.getElementById("msg").innerText="Submitted!";
return false;
}
```

```
/* GALLERY */
```

```
let images=[
  "https://picsum.photos/300?1",
  "https://picsum.photos/300?2",
  "https://picsum.photos/300?3"
];
```

```
let i=0;
```

```
function show(){
  document.getElementById("img").src=images[i];
}
```

```
function next(){
  i=(i+1)%images.length;
  show();
}
```

```
function prev(){
  i=(i-1+images.length)%images.length;
  show();
}
```

```
show();
```

```
/* BANNER */
```

```
function closeBanner(){
  document.getElementById("banner").style.display="none";
}
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

```
""""
```

```
@app.route('/')
```

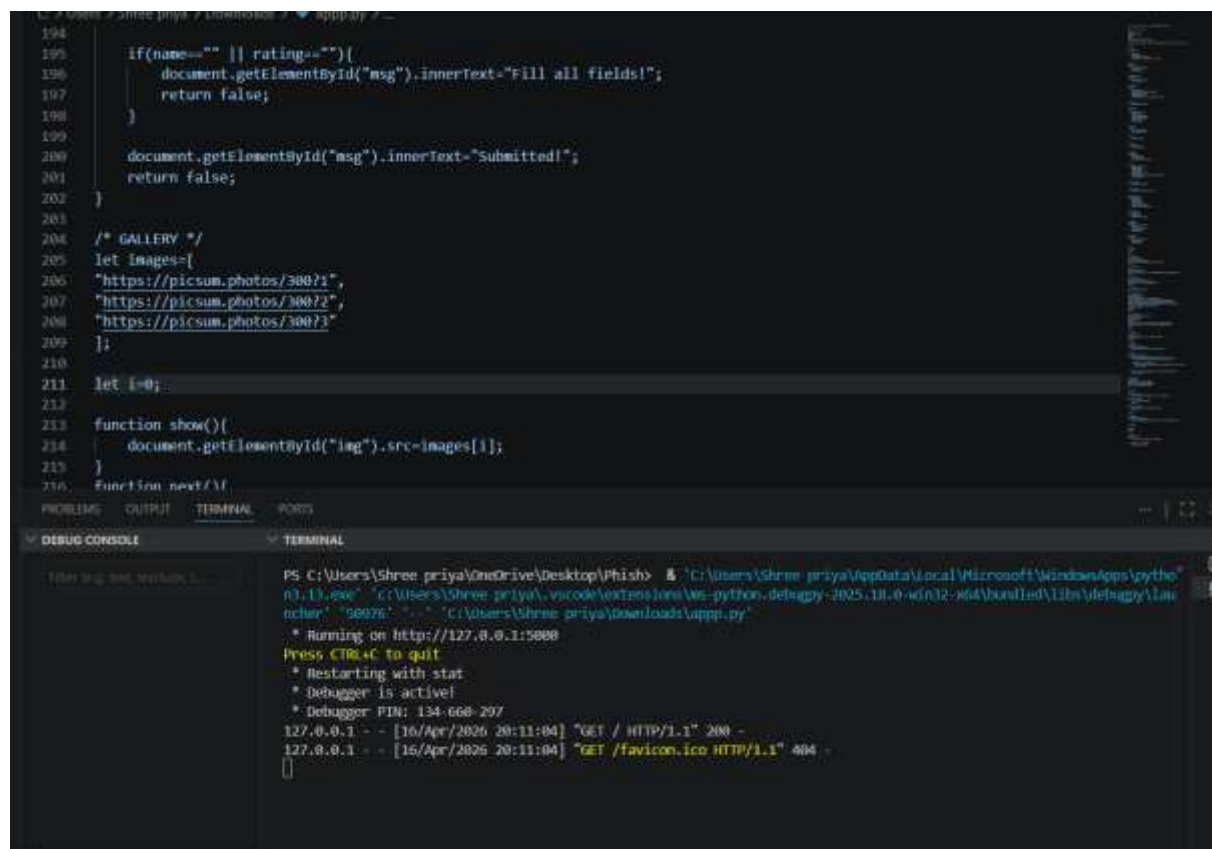
```
def home():
```

```
    return render_template_string(html)
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```

OUTPUT:

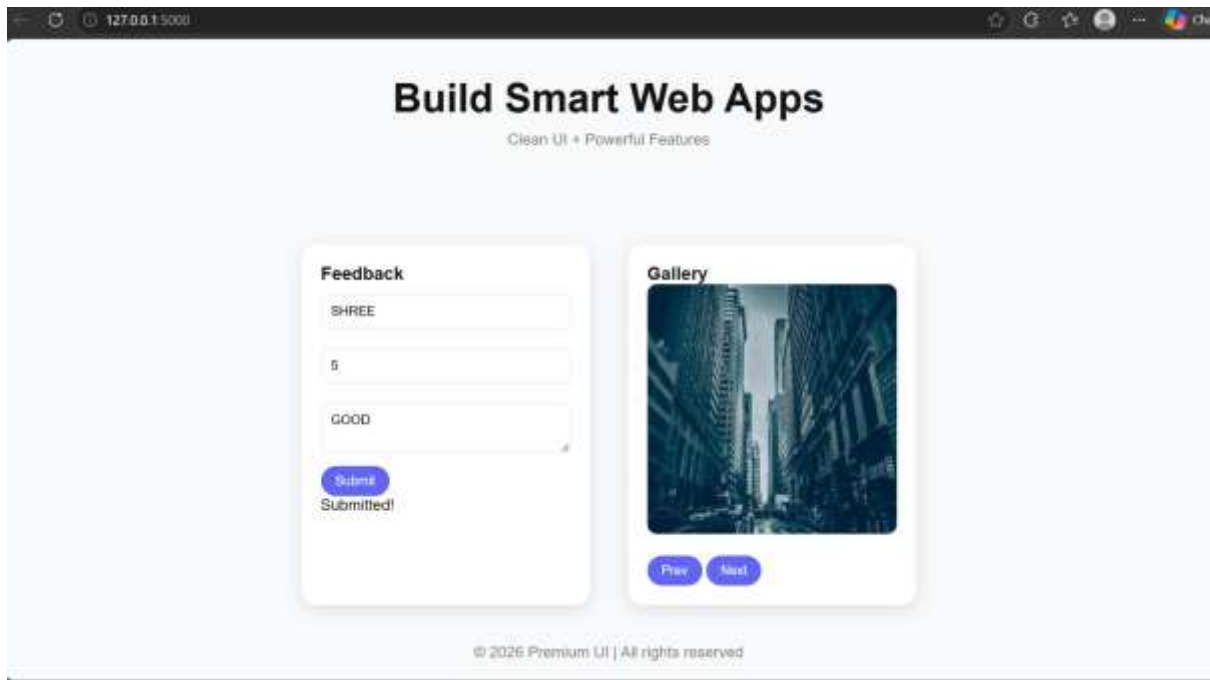


The screenshot shows a VS Code editor with a Python file named `app.py`. The code defines a Flask application with a single route `/` that renders an HTML template. The template includes a form with a text input and a submit button, and a gallery of three placeholder images from Picsum. The application is run in debug mode.

```
194
195     if(name==" || rating==" ){
196         document.getElementById("msg").innerText="Fill all fields!";
197         return false;
198     }
199
200     document.getElementById("msg").innerText="Submitted!";
201     return false;
202 }
203
204 /* GALLERY */
205 let images=[
206     "https://picsum.photos/300?1",
207     "https://picsum.photos/300?2",
208     "https://picsum.photos/300?3"
209 ];
210
211 let i=0;
212
213 function show(){
214     document.getElementById("img").src=images[i];
215 }
216
217 function next(){
```

The terminal output shows the command to run the application and the resulting output:

```
PS C:\Users\Shree priya\OneDrive\Desktop\Phish> & 'C:\Users\Shree priya\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'C:\Users\Shree priya\vscode\extensions\ms-python.debugpy-2025.11.0-win32-x64\bin\lib\debugpy\launcher' '50076' '-c' 'C:\Users\Shree priya\Downloads\app.py'
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PID: 134-668-297
127.0.0.1 - - [16/Apr/2026 20:11:04] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [16/Apr/2026 20:11:04] "GET /favicon.ico HTTP/1.1" 404 -
```



JUSTIFICATION

The design follows a **responsive and modern UI structure** using cards and flexible layouts.

JavaScript enhances **interactivity** through theme toggle, form validation, and gallery navigation.

CSS styling improves **user experience** with clean typography, spacing, and visual hierarchy.

The application demonstrates **real-world usability** by combining multiple UI components in one page.